Lectures

DATA 422: Advanced Data Science Methods and Ethics

Lectures

Below each week is a **course path** (what to do that week) and **learning objectives** for each lecture — what you should be able to do when that class ends. Full notes will land on the linked pages; until then, use the textbook sections listed there.

Textbook: Andriy Burkov, *The Hundred-Page Language Models Book* (free). Optional clips: Karpathy Zero-to-Hero. Labs are Colab notebooks. Friday quizzes are **paper, in class**; study banks are under Quizzes.

Week 1 — Contracts, loss, and tokens

Aug 24–28. Project 1 starts; Model 1 is not a premium.

Course path

Step	Material
Mon	Week 1 Monday — syllabus
Lab	Lab 1 · Open in Colab
Wed	Week 1 Wednesday — wildfire contract + Ch. 1
Fri	Week 1 Friday — Ch. 2 · **paper Quiz 1**
Study	Quiz 1 bank
Stone	Project 1 week 1 (due Wed Sep 2, 11:59 p.m.)

Monday — Syllabus and course path

24 Aug · essential · no LM-book chapter yet

- Find the syllabus, schedule, lab Colab, quiz study bank, and Project 1 stepping stone.
- Distinguish **ungraded Colab lab**, **paper Friday quiz**, and **project stepping stone**.
- State the grade weights (quizzes 40%; three projects 20% each) and that there is no written final.
- Open Google Colab with a Humboldt account and name CPU vs GPU for these labs.

Wednesday — Wildfire contract and Chapter 1

26 Aug · essential · LM book Ch. 1

- Distinguish **hazard**, **risk**, **observed damage**, and a **premium** or coverage decision.
- State the Model 1 target as $\mathbb{P}(\texttt{serious\_damage} = 1 \mid \text{inspected in/near a fire perimeter})$ and name what that probability is **not**.
- Define z (logit), $\hat{y}$, y, and a scalar **loss**; say why we update opposite the gradient.
- Put `zero_grad`, forward, `backward`, and `step` in order and say which line writes `.grad`.
- Explain why street, APN, and lat/lon are stripped from the student table.

Friday — Tokens, embeddings, likelihood

28 Aug · essential · LM book Ch. 2 · paper Quiz 1 (Ch. 1 + Lab 1)

- Contrast a **one-hot** vector with a learned **embedding**.
- Interpret cosine similarity near 1, 0, and -1.
- Describe **tokenization** and why the tokenizer changes sequence length T.
- State what a bigram / next-token model is estimating and why **lower NLL** is better on the same data.